# RMDL Assessment 2: Data Analysis Report

B282928

## 1 Background

In English, word frequency is among the most important predictors of lexical recognition reaction times, accounting for substantial variance beyond other word features (Brysbaert et al, 2011). However, word frequency facilitation of lexical recognition is not uniform among speaker populations of a language. Diependaele et al (2013) found that the word frequency facilitation effect was much steeper in L2 English speakers than native speakers. Diependaele et al describes this asymmetry with the Lexical Entrenchment Account, which predicts that L2 speakers have weaker lexical representations for low-frequency words, while native speakers maintain relatively stronger representations of low-frequency words in the lexicon. Under this account, L2 speakers show longer reaction times on average due to weaker lexical entrenchment. However, for L2 speakers, increases in word frequency lead to proportionally greater facilitation of lexical recognition than for native speakers. This finding has been replicated in lexical decision tasks in Brysbaert et al (2017), which found that the frequency and language group interaction described above disappeared when controlling for vocabulary size, suggesting that it is lexical entrenchment, and not simply L2 status, that drives the asymmetry.

While the asymmetric relationship between word frequency and word recognition reaction times between L1 and L2 speakers has been studied under the Lexical Entrenchment Account, the theory's application to semantic representations of the word forms is less clear. The Stroop paradigm assesses participants' ability to "inhibit cognitive interference that occurs when the processing of a specific stimulus feature [lexical orthography] impedes the simultaneous processing of a second stimulus attribute [color]," causing a reaction time delay to conflicting orthographic-color pairs as the semantic activation of the written word conflicts with the font color itself (Scarpina & Tagini, 2017; Stroop, 1935). Therfore, whereas Diependale et al (2013) and Brysbaert et al (2017) investigated lexical recognition, this study seeks to better understand the relationship between word frequency effects and automatic semantic interference between L1 and L2 English speakers through the Stroop task paradigm.

### 1.1 Research Questions

The Lexical Entrenchment Account predicts lower lexical entrenchment for L2 speakers of English overall, and a greater relative difference in entrenchment between frequent and infrequent English words.

- Does reduced lexical entrenchment for infrequent words in L2 speakers lead to a reduction in automatic semantic activation as seen through the Stroop effect?
- Word frequency has been observed to have an asymmetric impact between L1 and L2 speakers' lexical decision reaction times. Is a similar relationship observed between word frequency and automatic semantic interference in the Stroop effect?

### 1.2 Research Hypotheses

Following assumptions of the lexical entrenchment account,

1. I predict that L1 speakers of English will have similar lexical entrenchment for infrequent (rare) and frequent (common) word colors. This will lead to credibly longer reaction times between the color and non-color words, demonstrating the Stroop effect, but reaction times between rare and common collors will not be credibly statistically different.
2. On the other hand, L2 English speakers will have much lower lexical entrenchment for rare words in English, leading to reduced automatic semantic interference during the Stroop task. Thus, L2 participants will respond more quickly overall, and will display a credibly greater difference in reaction times between rare and common words due to the greater difference in lexical entrenchment between the two conditions as compared to L1 speakers.

## 2 Methods

In order to address the main research questions and hypothesis, this study uses a Stroop task paradigm (Stroop, 1935). Participants are presented with English color and non-color words of both high and low frequency, using Zipf score from the SUBTLEX UK database as a correlate of frequency. The stimuli set includes 16 words: 4 common colors, 4 rare colors, 4 common non-colors, and 4 rare non-colors. The non-color words are included as controls. A table including the total set of stimuli can be seen in Appendix A. The experiment varied all stimulus words using four font colors: black, blue, green, and brown, matching the common colors in the stimuli set. For the purposes of the Stroop task, common colors were judged as congruent and incongruent based on relation to the font color; all other stimuli types were judged as "control", as the congruence condition is not well defined.

Given that this is a reading task, the current study controls for length of stimulus words in letters by balancing word length across categories. Stimuli were orthogonally matched. Both common and rare conditions contain an identical distribution of word lengths, ranging from three to six letters with a mean length of 4.75 letters for color terms and 4.0 letters for non-color controls.

Stimuli were categorized into the "common" or "rare" condition based on their score on the Zipf scale from 1-7. These values were taken from the SUBTLEX Zipf (UK) data provided by the LexOPS dataset from Taylor et al (2020). All "common" stimulus items are Zipf scale value 4.8 or above, whereas all "rare" stimulus items are Zipf scale value 3.0 or below.

In total, seven people participated in the experiment. Three were native Standard Mandarin speakers, while four were native English speakers.

At the start of the experiment, participants were presented with the conditions of their participation in the study and asked for informed consent. They were then asked to fill out a language proficiency self-rating form describing their language proficiency in English and in Standard Mandarin with the following labels:

- Native Speaker
- High Proficiency (CEFR C1-C2)
- Intermediate Proficiency (CEFR B1-B2)
- Low Proficiency (CEFR A1-A2)
- No proficiency

An additional freeform text box was included for participants to indicate proficiency in any other languages. Full survey data concerning participant language proficiency self-reporting is shown in Appendix B.

After the language proficiency survey, an information screen was presented describing the experimental task. Participants were instructed that words would appear on the screen, and asked to indicate the font color of the text, ignoring the meaning of the word. They were asked to use the keyboard keys 1, 2, 3, and 4 to indicate the font color of the word, and to answer as accurately and as quickly as possible. Throughout the experiment, a legend displaying the key from number key to color was shown at the bottom of the screen: "Press 1 for BLACK, 2 for BLUE, 3 for GREEN, and 4 for BROWN."

Following this, eight practice trials were presented to the participant with two congruent, two incongruent, and four control trials. After the practice block, one main trial block was presented to participants. Every stimulus word was presented in all four font colors, in a fully randomized order, for a total of 96 trials. Reaction times for each trial was measured as the difference between stimulus presentation onset and participant keypress onset.

## 2.1 Statistical Analysis

Since this study investigates the difference in the inhibitory Stroop effect when color word and font color are in conflict, all "congruent" trials where the color word matched font color in common words were dropped from the data.

As advocated in Barr et al (2013), this study employs a maximal random effects structure to analyze experiment data, using a Bayesian linear random effects model from the `brms` package in R to assess the difference in frequency effects between English L1 and L2 participants:

$$ReactionTime \sim LanguageGroup * Common\_Rare * Color\_Noncolor + WordLength + (Common\_Rare + Color\_Noncolor + Length|Participant)$$

The primary dependent variable is reaction time, modeled using a LogNormal family. This structure explicitly models the interaction between group (L1 vs. L2), frequency of the stimulus word, and type (color word versus non-color word) and stimulus word length on reaction times, while controlling for word length's impact on reaction times. Given that all participants see multiple levels of word frequency and and type (color versus non-color word), these variables are included as random slopes.

To address the research hypotheses, several data and model parameters are of interest.

1a. L1 (Native) English speakers will have credibly longer reaction times in the common color condition as compared to the common non-color condition, replicating the Stroop effect.

- To evaluate this, I will look at the L1 group reaction times between common color words and common non-color words. If the Stroop effect is replicated, reaction time to name the font color should be credibly lower in the non-color word condition as compared to the color word condition.

1b. L1 (Native) English speakers will **not** have credibly longer reaction times between rare and common items within the color condition.

- To evaluate this, I look at L1 group reaction times between rare and common color words. If the lexical entrenchment account holds, reaction times from L1 English speakers between rare color words and common color words should not be credibly different.

2a. Compared to the L1 group, L2 (Non-native) English speakers will respond to all color items more quickly overall due to lower lexical entrenchment and thus less semantic interference when naming font colors.

- To assess this hypothesis, we will aggregate color word responses across the common and rare condition between L1 English and L2 English speakers. If the hypothesis holds, L2 English speakers will respond more quickly to all color words.

2b. Compared to the L1 group, L2 participants will demonstrate greater relative change in reaction times between rare and common items in the color condition, demonstrating the greater range of lexical entrenchment between the two levels of frequency.

- To look at this part of the hypothesis, we will look at overall reaction times in the color condition for both L1 English and L2 English speakers, comparing between the rare and common conditions. If the hypothesis holds, the range between rare and common condition reaction times should be greater overall in the L2 English group.

### 2.2 Data Availability Statement

All items relevant to the project are available on GitHub at **https://github.com/owenbraymond/rmdl-B282928**.

Anonymized participant experimental data used for this analysis can be found in the `/results/` folder of this repository.

The stimuli used in the experiment are generated within this file in Appendix A and saved as a .csv in `testable_materials` in this repository.

The Testable experiment file used to run the experiment is generated within this file in Appendix C and saved as a .csv in `testable_materials` in this repository. (This trial file is compatible with Testable as of April 29, 2026.)

## 3 Data

### 3.1 Workspace Setup, Package loading

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(languageR)
library(ggplot2)
library(devtools)
```

```
Loading required package: usethis
```

```r
library(tidybayes)
library(brms)
```

```
Loading required package: Rcpp
Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').

Attaching package: 'brms'

The following objects are masked from 'package:tidybayes':

    dstudent_t, pstudent_t, qstudent_t, rstudent_t

The following object is masked from 'package:stats':

    ar
```

```r
library(dplyr)
library(plotly)
```

```
Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

    last_plot

The following object is masked from 'package:stats':

    filter

The following object is masked from 'package:graphics':

    layout
```

```r
library(posterior)
```

```
This is posterior version 1.7.0

Attaching package: 'posterior'

The following objects are masked from 'package:stats':

    mad, sd, var

The following objects are masked from 'package:base':

    %in%, match
```

```r
library(kableExtra)
```

```

Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

    group_rows
```

```r
devtools::install_github("JackEdTaylor/LexOPS@*release")
```

```
Warning: `install_github()` was deprecated in devtools 2.5.0.
i Please use pak::pak("user/repo") instead.

Skipping install of 'LexOPS' from a github remote, the SHA1 (570ea099) has not changed since
  Use `force = TRUE` to force installation
```

```r
library(LexOPS)
```

## 3.2 Read and process the data

### 3.2.1 Importing Trial Data

First, I downloaded all participant experimental result files from Testable as individual .csv files and put them in my repository's "results" folder. I then used the `list.files()` function to create a list of all results files in the /results/ folder. using `lapply()`, I read all the .csv folders into individual tibbles within data_lst, skipping the first 3 rows which contained Testable metadata. I then used `bind_rows()` to combine all trial files into one tibble called `rmdl_data`, and wrote this into my repository's /data/ folder for use to any other interested researchers who want to start analysis with a combined results file.

```
files_to_read = list.files(
  path = "./results/",
  pattern = "*.csv$",
  recursive = TRUE,
  full.names = TRUE
)
data_lst <- lapply(files_to_read, function(x) {
  read_csv(x, skip = 3) #testable includes 3 rows at the top with metadata, skip these to ge
})
```

```
Rows: 107 Columns: 35
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (20): type, stim1, color, condition1, condition2, feedback, trialText, s...
dbl (15): rowNo, key, ITI, random, required, Zipf.SUBTLEX_UK, Length, 1:2, I...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 107 Columns: 35
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (20): type, stim1, color, condition1, condition2, feedback, trialText, s...
dbl (15): rowNo, key, ITI, random, required, Zipf.SUBTLEX_UK, Length, 1:2, I...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 107 Columns: 35
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (20): type, stim1, color, condition1, condition2, feedback, trialText, s...
dbl (15): rowNo, key, ITI, random, required, Zipf.SUBTLEX_UK, Length, 1:2, I...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 107 Columns: 35
-- Column specification --------------------------------------------------------
```

```
Delimiter: ","
chr (20): type, stim1, color, condition1, condition2, feedback, trialText, s...
dbl (15): rowNo, key, ITI, random, required, Zipf.SUBTLEX_UK, Length, 1:2, I...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 107 Columns: 35
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (20): type, stim1, color, condition1, condition2, feedback, trialText, s...
dbl (15): rowNo, key, ITI, random, required, Zipf.SUBTLEX_UK, Length, 1:2, I...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 107 Columns: 35
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (20): type, stim1, color, condition1, condition2, feedback, trialText, s...
dbl (15): rowNo, key, ITI, random, required, Zipf.SUBTLEX_UK, Length, 1:2, I...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 107 Columns: 35
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (20): type, stim1, color, condition1, condition2, feedback, trialText, s...
dbl (15): rowNo, key, ITI, random, required, Zipf.SUBTLEX_UK, Length, 1:2, I...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
rmdl_data <- bind_rows(data_lst, .id = "participant_num") %>%
  filter(condition2 != "congruent" | is.na(condition2)) #Since we are studying the inhibitory

write.csv(rmdl_data, "data/rmdl_data.csv")
```

## 3.3 Relevant Columns

`participant_num` : Unique participant ID for each participant.

`stim1` : The word text of the stimulus item.

`color` : The font color of the stimulus item.

`RT` : Participant reaction time to the stimulus item.

`rarity` : Whether the stimulus item belongs to the "rare" or "common" condition.

`color_noncolor` : Whether the stimulus item belongs to the "color" or "non-color" condition.

`Length` : Stimulus item length in letters.

`group` : Participant group, either "Mandarin" or "English", representing the participant's L1.

### 3.4 Splitting the Data

In the combined experimental data tibble, I used the `type` column to create two sub–tibbles, one for participant language proficiency self-rating information (`survey_responses`), and one for the reaction time data to stimulus items (`rmdl_data_RT`).

```
survey_responses <- rmdl_data %>%
  filter(type == "form") %>%
  mutate(
    english_level  = ifelse(head == "What is your English language READING ability?", respons
    mandarin_level = ifelse(head == "What is your Standard Mandarin (  ) language READING abi
    other_level    = ifelse(head == "Are you proficient in or learning any other foreign lang
  ) %>%
  group_by(participant_num) %>%
  summarise(
    english_level  = first(na.omit(english_level)),
    mandarin_level = first(na.omit(mandarin_level)),
    other_level    = first(na.omit(other_level))
  )

rmdl_data <- rmdl_data %>%
  left_join(survey_responses, by = "participant_num")

#creating labels for stats analysis
rmdl_data <- rmdl_data %>%
  mutate(
    group = case_when(
      english_level == "Native Speaker" ~ "English",
      mandarin_level == "Native Speaker" ~ "Mandarin",
      TRUE ~ "Other L2"
    )
  )

rmdl_data_RT <- rmdl_data %>%
```

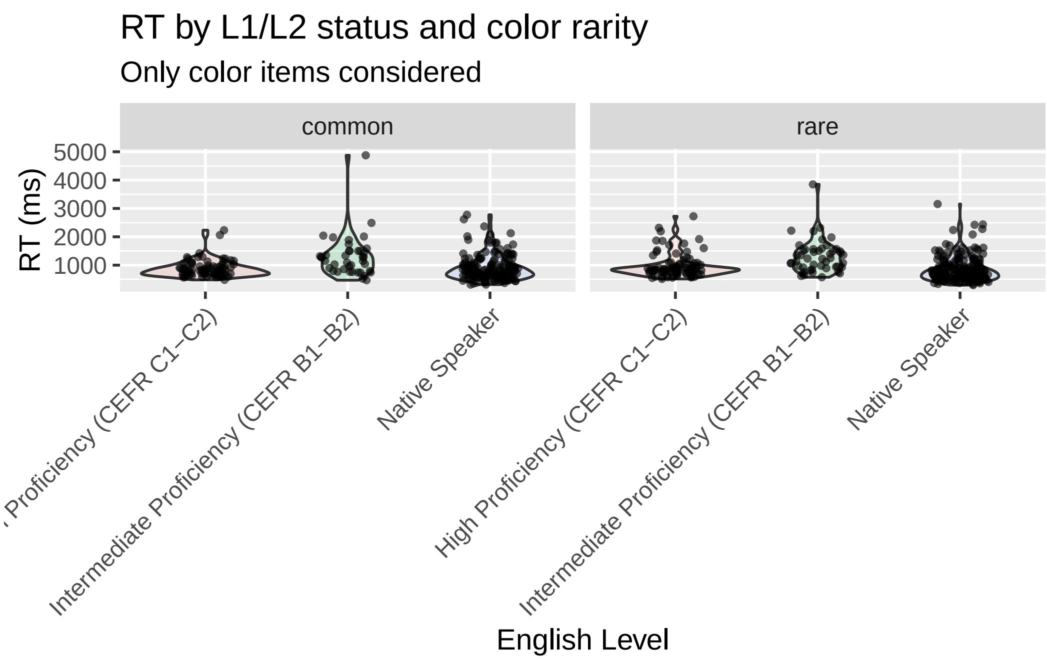
```
  #We will drop
  filter(type == "test" & condition1 != "practice") %>%
  select(participant_num, type, stim1, color, key, condition1, condition2, RT, english_level
```

## 3.5 Plotting the data

### 3.5.1 Figure 1.

```
#The code will print the figures, but you can find a saved vector graphic version in the 'fig
rawdata_violin <- rmdl_data_RT %>%
  ggplot(aes(x = english_level, y = RT, fill = english_level)) +
  geom_violin(alpha = 0.15) +
  geom_jitter(width = 0.2, alpha = 0.6, size = 0.8) +
  facet_wrap(~rarity) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position="none") +
  labs(x = "English Level", y = "RT (ms)", title = "RT by L1/L2 status and color rarity", sul

rawdata_violin
```



```
ggsave("figures/rawdata_violin.svg")
```

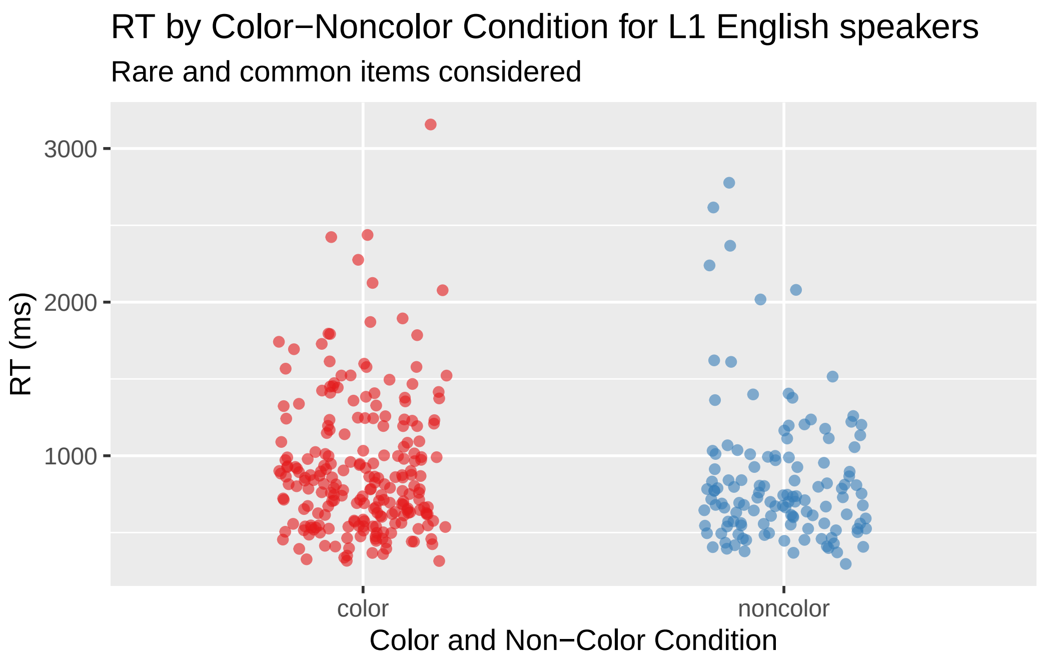
```
Saving 5.5 x 3.5 in image
```

Figure 1. This figure plots reaction times on the Y axis and groups English proficiency level of participants on the X axis, shown in red, green, and blue for visual clarity. This plot shows that while there was variation between group reaction times, patterns look fairly similar between the common and rare conditions overall.

#### 3.5.2 Figure 2.

```
rawdata_L1_color_noncolor_condition <- rmdl_data_RT %>%
  filter(group == "English") %>%
  ggplot(aes(x = color_noncolor, y = RT, color=color_noncolor)) +
  geom_jitter(width = 0.2, alpha = 0.6, size = 1.4) +
  scale_color_brewer(name = "Color-Noncolor Condition", labels = c("Color", "Noncolor"), pale
  labs(x = "Color and Non-Color Condition", y = "RT (ms)", title = "RT by Color-Noncolor Cond

rawdata_L1_color_noncolor_condition
```



```
ggsave("figures/rawdata_L1_color_noncolor_condition.svg")
```

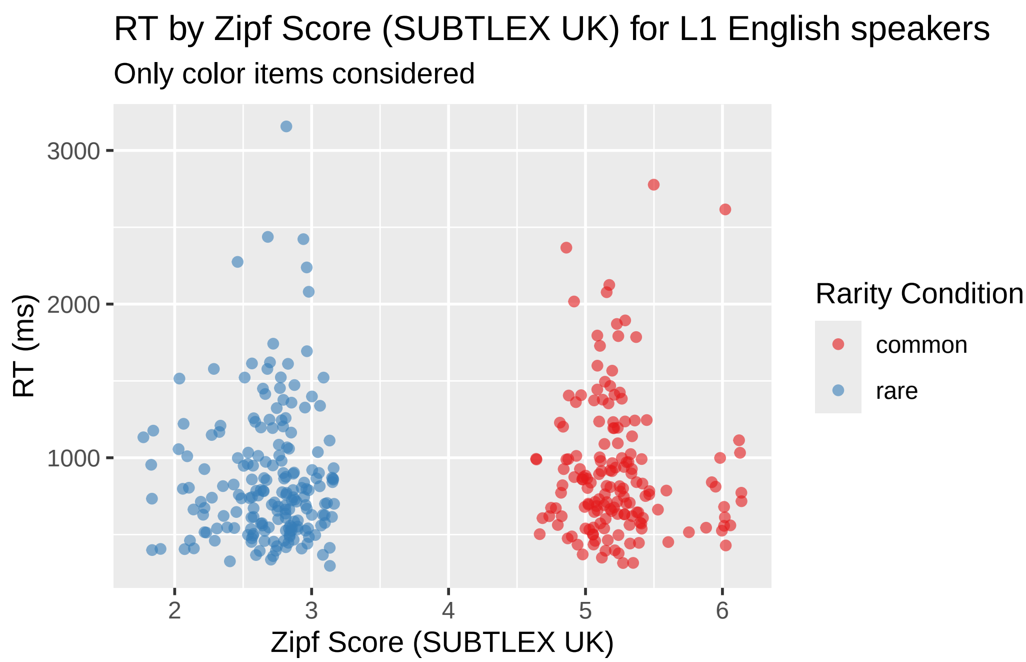
```
Saving 5.5 x 3.5 in image
```

Figure 2. This figure plots reaction times on the Y axis and the color-noncolor condition on the X axis. Only L1 English speakers are included in the plot, as this is a test of Hypothesis 1a; if L1 speakers of English are slower to respond to color words versus non-color words, there is evidence of replicating the Stroop effect. For visual clarity, color word items are shown in red and non-color word items are shown in blue. This plot indicates that L1 English speakers in this study may have had slightly slower reaction times to color words as compared to non-color words, but the robustness of this difference will be assessed in the statistical model.

### 3.5.3 Figure 3.

```
rawdata_L1_color_condition <- rmdl_data_RT %>%
  filter(group == "English") %>%
  ggplot(aes(x = Zipf.SUBTLEX_UK, y = RT, color=rarity)) +
  geom_jitter(width = 0.2, alpha = 0.6, size = 1.4) +
  scale_color_brewer(name = "Rarity Condition", labels = c("common", "rare"), palette = "Set
  labs(x = "Zipf Score (SUBTLEX UK)", y = "RT (ms)", title = "RT by Zipf Score (SUBTLEX UK) 

rawdata_L1_color_condition
```




```
ggsave("figures/rawdata_L1_color_condition.svg")
```

```
Saving 5.5 x 3.5 in image
```

Figure 3. This figure plots reaction times on the Y axis and Zipf score (SUBTLEX UK) as a correlate of rarity on the X axis. Only L1 English speakers are included in the plot, as this is a test of Hypothesis 1b; if L1 speakers of English have similar lexical entrenchment of rare and common color words, reaction times to both conditions should be similar. For visual clarity, common items are shown in red and rare items are shown in blue. This plot indicates that there may not be significantly different reaction times between common and rare colors among L1 speakers of English in this study.

### 3.6 Summarize the data

```
rmdl_data_RT %>%
  group_by(group) %>%
  summarise(mean = mean(RT), median(RT), sd(RT), min(RT), max(RT), n = n())
```

```
# A tibble: 2 x 7
  group      mean `median(RT)` `sd(RT)` `min(RT)` `max(RT)`     n
```

```
  <chr>    <dbl>         <dbl>    <dbl>     <dbl>     <dbl> <int>
1 English   886.          778.     443.       296      3156   352
2 Mandarin 1049.          890      508.       469      4877   264
```

## 4 Regression modelling

To model this data, I chose a LogNormal family. Reaction time data is often right-skewed, meaning it has a long tail to the right. Modeling with a LogNormal family "squishes" the data, allowing the model to interpret it as if it were close to a Gaussian distribution.

As for model parameters, this model explicitly seeks to understand the interaction between language group (L1 versus L2 English), stimulus item rarity (common versus rare), and color versus non-color word. This addresses the core hypotheses of the study, assessing L1 and L2 interactions with word frequency (rarity) and status as a color word. The model includes word length in letters to control for the reading effect that may be incurred from longer or shorter word length.

The model employs a maximal random effects model, including random slopes for each participant for word rarity, color-noncolor status, and word length. These random slopes were chosen as these were the three within-participant conditions that all participants experienced at different levels.

```r
my_seed = 4423

rmdl_data_bm <- brm(
  RT ~ group * rarity * color_noncolor + Length + (rarity + color_noncolor + Length | partic
  data = rmdl_data_RT,
  family = lognormal,
  iter = 4000,
  cores = 4,
  seed = my_seed,
  file = "cache/rmdl_bm",
  control = list(adapt_delta = 0.99, max_treedepth = 15)
  )
```

**Model Summary:**

```r
summary(rmdl_data_bm, prob = 0.95)
```

```
 Family: lognormal
  Links: mu = identity
```

```
Formula: RT ~ group * rarity * color_noncolor + Length + (rarity + color_noncolor + Length |
   Data: rmdl_data_RT (Number of observations: 616)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~participant_num (Number of levels: 7)
                                       Estimate Est.Error l-95% CI u-95% CI
sd(Intercept)                              0.42      0.25     0.09     1.02
sd(rarityrare)                             0.06      0.06     0.00     0.22
sd(color_noncolornoncolor)                 0.08      0.08     0.00     0.27
sd(Length)                                 0.03      0.03     0.00     0.11
cor(Intercept,rarityrare)                 -0.01      0.44    -0.81     0.81
cor(Intercept,color_noncolornoncolor)     -0.08      0.42    -0.82     0.73
cor(rarityrare,color_noncolornoncolor)     0.10      0.45    -0.77     0.86
cor(Intercept,Length)                     -0.05      0.45    -0.84     0.80
cor(rarityrare,Length)                    -0.01      0.46    -0.83     0.82
cor(color_noncolornoncolor,Length)         0.02      0.45    -0.81     0.82
                                       Rhat Bulk_ESS Tail_ESS
sd(Intercept)                          1.00     2473     2249
sd(rarityrare)                         1.00     3106     3998
sd(color_noncolornoncolor)             1.00     2484     2668
sd(Length)                             1.00     3145     3274
cor(Intercept,rarityrare)              1.00     7438     4890
cor(Intercept,color_noncolornoncolor)  1.00     8150     5765
cor(rarityrare,color_noncolornoncolor) 1.00     5082     5537
cor(Intercept,Length)                  1.00     8779     4999
cor(rarityrare,Length)                 1.00     7085     6281
cor(color_noncolornoncolor,Length)     1.00     6627     6552

Regression Coefficients:
                                         Estimate Est.Error l-95% CI
Intercept                                    6.79      0.27     6.25
groupMandarin                                0.12      0.36    -0.61
rarityrare                                  -0.09      0.06    -0.22
color_noncolornoncolor                      -0.14      0.08    -0.30
Length                                      -0.01      0.03    -0.06
groupMandarin:rarityrare                     0.10      0.10    -0.09
groupMandarin:color_noncolornoncolor         0.06      0.12    -0.16
rarityrare:color_noncolornoncolor            0.08      0.07    -0.07
groupMandarin:rarityrare:color_noncolornoncolor     0.07      0.11    -0.16
                                         u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept                                    7.33 1.00     3328     3320
```

```
groupMandarin                                   0.87 1.00     3170     3123
rarityrare                                      0.03 1.00     4256     3940
color_noncolornoncolor                          0.02 1.00     3483     3706
Length                                          0.05 1.00     6970     5471
groupMandarin:rarityrare                        0.28 1.00     4097     4086
groupMandarin:color_noncolornoncolor            0.30 1.00     3346     3306
rarityrare:color_noncolornoncolor               0.22 1.00     5428     6056
groupMandarin:rarityrare:color_noncolornoncolor     0.29 1.00     5262     6287

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.34      0.01     0.32     0.36 1.00    12332     6122

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

### 4.1 Looking at MCMC draws from the model

```
rmdl_data_bm_draws <- as_draws_df(rmdl_data_bm)

# Now, following the QML textbook lesson 35, we transform the predictors relative relationshi
rmdl_data_bm_draws_combined <- rmdl_data_bm_draws %>% mutate(
  eng_common_color = b_Intercept,
  mand_common_color = b_Intercept + b_groupMandarin,
  eng_rare_color = b_Intercept + b_rarityrare,
  eng_common_noncolor = b_Intercept + b_color_noncolornoncolor,
  mand_rare_color = b_Intercept + b_rarityrare + `b_groupMandarin:rarityrare`,
  mand_common_noncolor = b_Intercept + b_color_noncolornoncolor + `b_groupMandarin:color_nonc
  eng_rare_noncolor = b_Intercept + b_rarityrare + `b_rarityrare:color_noncolornoncolor`,
  mand_rare_noncolor = b_Intercept + b_groupMandarin + b_rarityrare + `b_rarityrare:color_non
) %>%
  select(eng_common_color, mand_common_color, eng_rare_color, eng_common_noncolor, mand_rare_
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

Now, we can look at the relevant variables described in Methods to address the research hypothesis.

### 4.1.1 Figure 5. L1 Stroop Replication

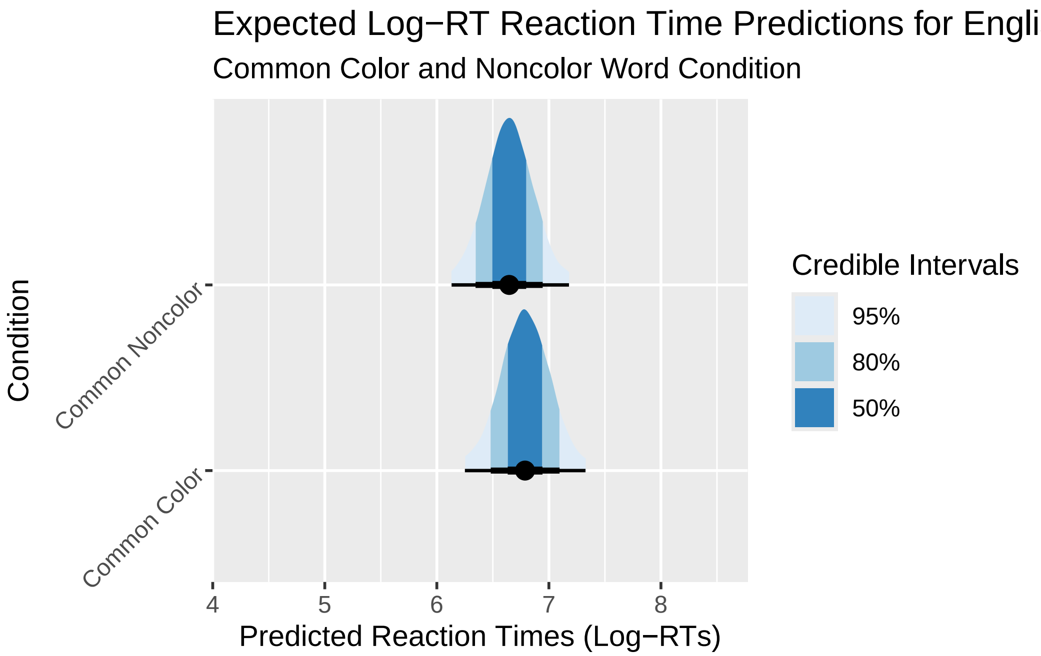
For hypothesis 1a, L1 reaction times in the color condition should be credibly slower than in the noncolor condition, regardless of rarity.

I am using the QML textbook Chapter 35 instructions to create a plot to assess the difference between these conditions.

```
draws_combined_epred_H1 <- rmdl_data_bm_draws_combined %>%
  select(eng_common_color, eng_common_noncolor) %>%
  pivot_longer(eng_common_color:eng_common_noncolor, names_to = "cond", values_to = "epred")
  mutate(
    cond = case_when(
      cond == "eng_common_color" ~ "Common Color",
      cond == "eng_common_noncolor" ~ "Common Noncolor"
    )
  )

H1_epreds_plot <- draws_combined_epred_H1 %>%
  ggplot(aes(epred, cond)) +
  stat_halfeye(
    .width = c(0.5, 0.8, 0.95),
    aes(fill = after_stat(level))
  ) +
  theme(axis.text.y = element_text(angle = 45, hjust = 1)) +
  scale_fill_brewer(na.translate = FALSE, name="Credible Intervals", labels = c("95%", "80%"
  labs(
    title="Expected Log-RT Reaction Time Predictions for English L1 Speakers",
    subtitle="Common Color and Noncolor Word Condition",
    x="Predicted Reaction Times (Log-RTs)",
    y="Condition")

H1_epreds_plot
```




```
ggsave("figures/H1_epreds_plot.svg")
```

```
Saving 5.5 x 3.5 in image
```

### 4.1.2 Figure 6. L1 Rare and Common Colors - Reaction Time Equivalence Hypothesis

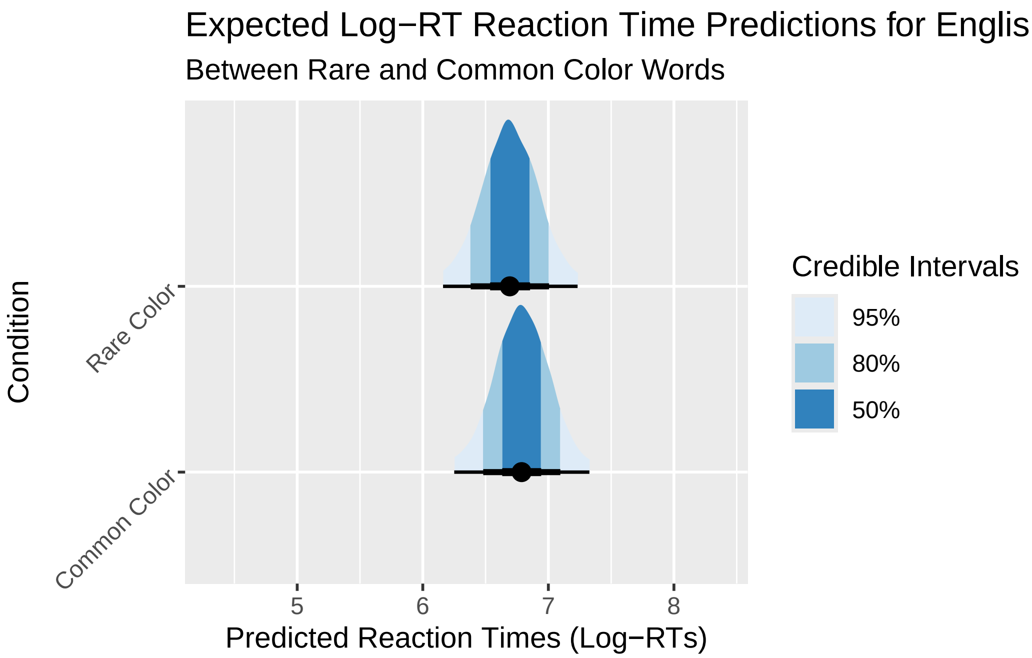
For hypothesis 1b, L1 reaction times should not be credibly different between the rare and common color conditions.

```
draws_combined_epred_H2 <- rmdl_data_bm_draws_combined %>%
  select(eng_common_color, eng_rare_color) %>%
  pivot_longer(eng_common_color:eng_rare_color, names_to = "cond", values_to = "epred") %>%
  mutate(
    cond = case_when(
      cond == "eng_common_color" ~ "Common Color",
      cond == "eng_rare_color" ~ "Rare Color",
      TRUE ~ "" #Catch-all, should not ever be used if everything is working
    )
  )

H2_epreds_plot <- draws_combined_epred_H2 %>%
```

```
  ggplot(aes(epred, cond)) +
  stat_halfeye(
    .width = c(0.5, 0.8, 0.95),
    aes(fill = after_stat(level))
  ) +
  theme(axis.text.y = element_text(angle = 45, hjust = 1)) +
  scale_fill_brewer(na.translate = FALSE, name="Credible Intervals", labels = c("95%", "80%"
  labs(title="Expected Log-RT Reaction Time Predictions for English L1 Speakers", subtitle="

H2_epreds_plot
```




```
ggsave("figures/H2_epreds_plot.svg")
```

```
Saving 5.5 x 3.5 in image
```

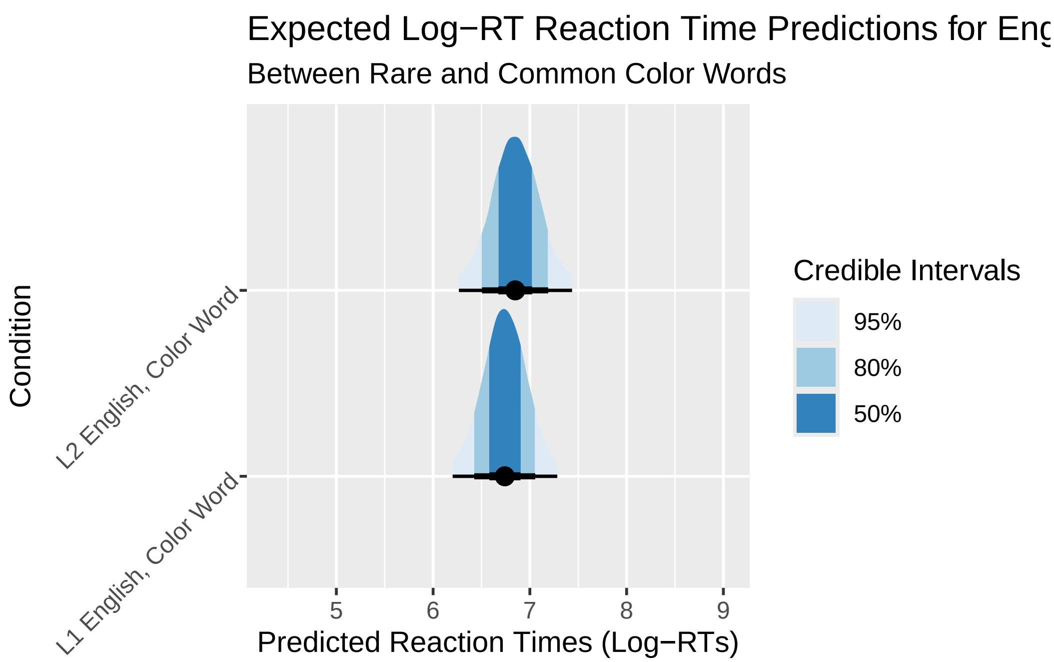
#### 4.1.3 Figure 7. L2 Faster Overall Reaction Times to Colors

For Hypothesis 2a, I assess the expected predictions for all L2 color responses versus all L1 color responses.

```
draws_combined_epred_H3 <- rmdl_data_bm_draws_combined %>%
  select(eng_common_color, mand_common_color, eng_rare_color, mand_rare_color) %>%
  pivot_longer(eng_common_color:mand_rare_color, names_to = "cond", values_to = "epred") %>%
  #Combining all color responses while keeping L1 and L2 separate to compare color responses
  mutate(
    cond = case_when(
      cond == "eng_common_color" | cond == "eng_rare_color" ~ "L1 English, Color Word",
      cond == "mand_common_color" | cond == "mand_rare_color" ~ "L2 English, Color Word",
      TRUE ~ "" #Catch-all, should not ever be used if everything is working
    )
  )

H3_epreds_plot <- draws_combined_epred_H3 %>%
  ggplot(aes(epred, cond)) +
  stat_halfeye(
    .width = c(0.5, 0.8, 0.95),
    aes(fill = after_stat(level))
  ) +
  theme(axis.text.y = element_text(angle = 45, hjust = 1)) +
  scale_fill_brewer(na.translate = FALSE, name="Credible Intervals", labels = c("95%", "80%"
  labs(title="Expected Log-RT Reaction Time Predictions for English L1 Speakers", subtitle="

H3_epreds_plot
```




```
ggsave("figures/H3_epreds_plot.svg")
```

```
Saving 5.5 x 3.5 in image
```

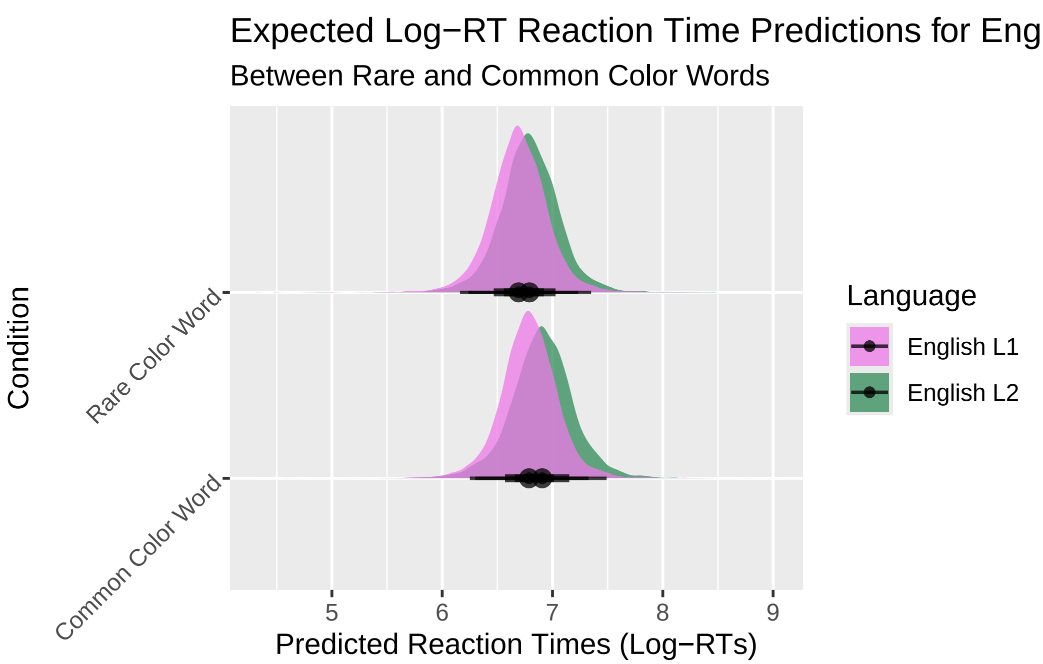
#### 4.1.4 Figure 8. L2 Greater Difference Between Rare and Common Color Reaction Times

Now we look at the statistical average differences between rare and common color RTs for L2 and L1 English speakers.

```
draws_combined_epred_H4 <- rmdl_data_bm_draws_combined %>%
  select(eng_common_color, mand_common_color, eng_rare_color, mand_rare_color) %>%
  pivot_longer(eng_common_color:mand_rare_color, names_to = "cond", values_to = "epred") %>%
  separate(cond, c("Language","Rarity")) %>%
  #Simple mutate to rename for plot Y axis labels
  mutate(
    Rarity = case_when(
      Rarity == "rare" ~ "Rare Color Word",
      Rarity == "common" ~ "Common Color Word",
      TRUE ~ "" #Catch-all, should not ever be used if everything is working
    )
  )
```

```
Warning: Expected 2 pieces. Additional pieces discarded in 32000 rows [1, 2, 3, 4, 5, 6,
7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, ...].
```

```
draws_combined_epred_H4 %>%
  ggplot(aes(epred, Rarity, fill=Language)) +
  stat_halfeye(alpha=0.75) +
  scale_fill_manual(labels = c("English L1", "English L2"), values = c("orchid2", "seagreen")
  theme(axis.text.y = element_text(angle = 45, hjust = 1)) +
  labs(title="Expected Log-RT Reaction Time Predictions for English L1 and L2 Speakers", sub
```




# 5 Discussion

*Report the results of the model. Write a brief discussion of the your model results in light of the research question.*

## 5.1 Limitations

Should have matched common/rare word actual color identity much more closely, if length could be matched too

# 6 Appendix

## 6.1 A. Stimuli Generation and Table

The actual trial will reduce to 4 test colors to allow participants to lay 4 fingers over the 1-4 keyboard keys, increasing viability of reaction times. 8 keys may be too slow in baseline response to generate useful data.

with that in mind, let's cut the tibble down to size while keeping the word length balance.

```
lexops_df <- LexOPS::lexops

common_colors <- c("green", "brown", "blue", "black")
rare_colors <- c("sepia", "ochre", "mauve", "aqua")
common_noncolors <- c("fast", "slow", "big", "small")
rare_noncolors <- c("apex", "tusk", "nymph", "irk")

my_words <- c(common_colors, rare_colors, common_noncolors, rare_noncolors)

word_df <- tibble(string = my_words) %>%
  mutate(rarity = case_when(
    string %in% common_colors | string %in% common_noncolors ~ "common",
    string %in% rare_colors | string %in% rare_noncolors ~ "rare",
    TRUE ~ "other" # Catch-all, should never be used
  ))

word_df <- word_df %>%
  mutate(color_noncolor = case_when(
    string %in% common_colors | string %in% rare_colors ~ "color",
    string %in% common_noncolors | string %in% rare_noncolors ~ "noncolor",
    TRUE ~ "other" # Catch-all, should never be used
  ))

lookup_table <- lexops_df %>%
  select(string, Zipf.SUBTLEX_UK, Length)

word_df_final <- word_df %>%
  left_join(lookup_table, by = "string")

word_df_final <- word_df_final |>
  mutate(Zipf.SUBTLEX_UK = round(Zipf.SUBTLEX_UK, digits=2)) |>
  mutate(combined_condition = paste(rarity, color_noncolor, sep = "_"))
```

```
word_df_final %>%
  select(string, rarity, color_noncolor, Zipf.SUBTLEX_UK, Length) %>%
  knitr::kable(
    digits = 2,
    col.names = c("Word", "Rarity", "Color or Noncolor", "SUBTLEX Zipf Score (UK)", "Length i
  ) %>% kable_styling()
```

| Word | Rarity | Color or Noncolor | SUBTLEX Zipf Score (UK) | Length in Letters |
|---|---|---|---|---|
| green | common | color | 5.22 | 5 |
| brown | common | color | 5.01 | 5 |
| blue | common | color | 5.16 | 4 |
| black | common | color | 5.26 | 5 |
| sepia | rare | color | 2.38 | 5 |
| ochre | rare | color | 2.77 | 5 |
| mauve | rare | color | 2.71 | 5 |
| aqua | rare | color | 2.98 | 4 |
| fast | common | noncolor | 5.09 | 4 |
| slow | common | noncolor | 4.81 | 4 |
| big | common | noncolor | 5.94 | 3 |
| small | common | noncolor | 5.43 | 5 |
| apex | rare | noncolor | 2.96 | 4 |
| tusk | rare | noncolor | 3.00 | 4 |
| nymph | rare | noncolor | 2.72 | 5 |
| irk | rare | noncolor | 1.95 | 3 |

```
write_csv(word_df_final, "testable_materials/exp_stimuli.csv")
```

## 6.2 B. Participant Language Proficiency Survey Results

```
survey_responses %>%
  knitr::kable(
    digits = 2,
    col.names = c("Participant ID", "English Proficiency", "Standard Mandarin Proficiency", "
  ) %>% kable_styling()
```

| Participant ID | English Proficiency | Standard Mandarin Proficiency | Other Language |
|---|---|---|---|
| 1 | Native Speaker | No proficiency | Spanish (C1) Fr |
| 2 | Native Speaker | No proficiency | Spanish A1 |
| 3 | Native Speaker | No proficiency | French - 8 Italia |
| 4 | High Proficiency (CEFR C1-C2) | Native Speaker | No |
| 5 | Intermediate Proficiency (CEFR B1-B2) | Native Speaker | NA |
| 6 | Native Speaker | No proficiency | Intermediate Fr |
| 7 | High Proficiency (CEFR C1-C2) | Native Speaker | NA |

## 6.3 C. Building the Testable file

We then create a "key" tibble to map keypress responses to font color.

```
keypress_key <- tibble(
  color = c("stim1: black","stim1: blue","stim1: green","stim1: brown"),
  key = c(1, 2, 3, 4)
)
```

We cross the conditions to get the experimental stimuli and keys.

```
exp_trialfile <- crossing(word_df_final, keypress_key)
```

Now, let's finalize the Testable trial file to label conditions as "congruent" (text color name is the same as font color), "incongruent" (text color name is NOT the same as font color), or "control" (text color name cannot be related to font color).

```
exp_trialfile <- exp_trialfile %>% mutate(
  congruence_condition = case_when(
    color_noncolor == "color" & rarity == "common" & paste0("stim1: ", string) == color ~ "co
    color_noncolor == "color" & rarity == "common" & paste0("stim1: ", string) != color ~ "in
    TRUE ~ "control"
  )
)
```

We want to get a 1:2 ratio of critical vs control trials

```
exp_trialfile_crit <- exp_trialfile %>%
  filter(color_noncolor == "color") %>%
  crossing(1:2)


exp_trialfile_cont <- exp_trialfile %>%
  filter(color_noncolor != "color")

exp_trialfile <- bind_rows(exp_trialfile_crit, exp_trialfile_cont)
```

Let's add in the instructions and practice trials for testable.

```
head_block1 <- tibble(
  type = "instructions",
  stim1 = "",
  color = "",
  key = 0,
  condition1 = "",
  condition2 = "",
  feedback = "",
  trialText = "",
  stimFormat = "",
  ITI   = 0,
  button1 = "NEXT",
  keyboard = "",
  random = 1,
  required = "",
  title = "Welcome to the experiment",
  content = "First, we'd like to collect some information about your language background. Ple
  head = "",
  responseType = ""
)

#English Proficiency Check Row
head_block2 <- tibble(
  type = "form",
  stim1 = "",
  color = "",
  key = 0,
  condition1 = "",
  condition2 = "",
  feedback = "",
```

```
  trialText = "",
  stimFormat = "",
  ITI   = 0,
  button1 = "NEXT",
  keyboard = "",
  random = 2,
  title = "",
  content = "",
  responseType = "radio",
  required = "1",
  head = "What is your English language READING ability?",
  responseOptions = "Native Speaker;High Proficiency (CEFR C1-C2);Intermediate Proficiency (
)

#Mandarin Proficiency Check Row
head_block3 <- tibble(
  type = "form",
  stim1 = "",
  color = "",
  key = 0,
  condition1 = "",
  condition2 = "",
  feedback = "",
  trialText = "",
  stimFormat = "",
  ITI   = 0,
  button1 = "",
  keyboard = "",
  random = 3,
  title = "",
  content = "",
  responseType = "radio",
  required = "1",
  head = "What is your Standard Mandarin (  ) language READING ability?",
  responseOptions = "Native Speaker;High Proficiency (CEFR C1-C2);Intermediate Proficiency (
)

head_block4 <- tibble(
  type = "form",
  stim1 = "",
  color = "",
  key = 0,
```

```
  condition1 = "",
  condition2 = "",
  feedback = "",
  trialText = "",
  stimFormat = "",
  ITI   = 0,
  button1 = "",
  keyboard = "",
  random = 4,
  title = "",
  content = "",
  responseType = "box",
  required = "",
  head = "Are you proficient in or learning any other foreign languages? (Please include the
  responseOptions = ""
)

head_block5 <- tibble(
  type = "instructions",
  stim1 = "",
  color = "",
  key = 0,
  condition1 = "",
  condition2 = "",
  feedback = "",
  trialText = "",
  stimFormat = "",
  ITI   = 0,
  button1 = "NEXT",
  keyboard = "",
  random = 5,
  required = "",
  title = "Thank you for your survey responses! ",
  content = "Let's move on to the experiment. \nIn this study, you are asked to indicate the
Press 1 for BLACK, <br>2 for <span style=\"color:blue;\">BLUE</span>, <br>3 for <span style=\

Please answer as accurately and as quickly as possible. Let's do some practice!",
  head = "",
  responseType = ""
)
```

```r
practice_block <- tribble(
  ~stim1,   ~color, ~key,
  "BLACK",  "stim1: black",    1,
  "GREEN",  "stim1: black",    1,
  "BLUE",   "stim1: blue",     2,
  "BROWN",  "stim1: green",    3,
  "TUSK",   "stim1: blue",     2,
  "SEPIA",  "stim1: green",    3,
  "IRK",    "stim1: brown",    4,
  "NYMPH",  "stim1: brown",    4
  ) %>%
  mutate(
    type = "test",
    condition1 = "practice",
    feedback = glue::glue('correct: Correct! You responded in %RT% ms ; incorrect: Incorrect
    stimFormat = "word",
    ITI = 400,
    keyboard = "1 2 3 4",
    random = 6,
    required = "",
    title = "",
    content = "",
    trialText = "Press 1 for BLACK, <br>2 for <span style=\"color:blue;\">BLUE</span>, <br>3
    button1 = "",
    head = "",
    responseType = ""
  )

head_block6 <- tibble(
  type = "instructions",
  stim1 = "",
  color = "",
  key = 0,
  condition1 = "",
  condition2 = "",
  feedback = "",
  trialText = "",
  stimFormat = "",
  ITI   = 0,
  button1 = "NEXT",
  keyboard = "",
  random = 7,
```

```
  required = "",
  title = "Ready to start?",
  content = "Remember: Press 1 for BLACK, <br>2 for <span style=\"color:blue;\">BLUE</span>,
  head = "",
  responseType = ""
)
```

Now we transform it into a Testable-usable file:

```
exp_trialfile_main <- exp_trialfile %>% mutate(
  type = "test",
  stim1 = toupper(string),
  condition1 = case_when(
    congruence_condition != "control" ~ "critical",
    congruence_condition == "control" ~ "filler"
  ),
  condition2 = congruence_condition,
  trialText = case_when(
    type != "instructions" ~ "Press 1 for BLACK, <br>2 for <span style=\"color:blue;\">BLUE<
    TRUE ~ ""
  ),
  feedback = "",
  stimFormat = case_when(
    type != "instructions" ~ "word",
    TRUE ~ ""
  ),
  ITI = case_when(
    type != "instructions" ~ 400,
    TRUE ~ 0
  ),
  button1 = case_when(
    type == "instructions" ~ "NEXT",
    TRUE ~ ""
  ),
  keyboard = case_when(
    type != "instructions" ~ "1 2 3 4",
    TRUE ~ ""
  ),
  random = 8,
  required = "",
  title = "",
  content = "",
```

```
  head = "",
  responseType = ""
)
```

Now let's save it for use in Testable.

```
exp_trialfile_final <- bind_rows(head_block1, head_block2, head_block3, head_block4, head_bl
  mutate(stimSize = "175%")

write_csv(exp_trialfile_final, "testable_materials/exp_trialfile.csv")
```